Source-Locked Production Registry Crosswalks: A Finite Exact-Matching and First-Mismatch Certificate

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Abstract

We audit a fixed source-locked corpus of nine distinct objects that could plausibly enter the production packet-prefix crosswalk left open by TPC-194. Every object is checked against seven production axes and three noninterchangeable formula types. There are no complete crosswalks. The first production-axis mismatch for every object is the absence of a source-locked named production atom. The exact verdict is `FIRST_MISMATCH_CERTIFIED_NOT_TESTABLE`. This is an L0/L1 finite-corpus result, not a fixed-atom cancellation theorem, a direct-route reopen, or an L2 gain.

1 Authorization and mathematical boundary

TPC-203 stopped with

`SOURCE_LOCKED_PRODUCTION_PACKET_PREFIX_CROSSWALK`

as the direct-production first missing node [10]. The user subsequently authorized one finite TPC-204 whose outcome had to be either an exact unique crosswalk or a certified first mismatch. That authorization is workflow input only. It neither supplies source data nor makes any reopen trigger true.

The target still requires a source-locked named physical atom, an exact packet schedule, one common $X/N/q$ range, a uniform constant C, a positive fixed-X exponent σ, one literal target normalization, and a complete physical-loss ledger. All seven fields must belong to the same production record. Explanatory prose may not fill a null field.

2 Declared candidate universe

The corpus is closed by explicit source bindings and canonical object selectors, not by a keyword scan. It contains the direct-production lineage objects isolated in TPC-159, TPC-167, TPC-180, TPC-183, TPC-184, TPC-189, TPC-193, TPC-194, and the TPC-203 import boundary [2, 3, 4, 5, 6, 7, 8, 9, 10]. Duplicate imports and auxiliary phase-measure, grid, and interval statements are recorded separately rather than silently discarded.

Canonical object	Native source object	First production mismatch
`H9.phase_cell_registry`	Empty source-locked registry slot	Named atom ID and phase locator are null
`TT26.RATIONAL_PERIODIC_ATOM`	q/N terminal block with log saving	Rational production atom value is absent
`A159.DYADIC_SHADOW_PREFIX`	q/T cumulative prefix off the shadow	Periodic ρ is not a named atom
`A167.DIRECT_TWIST_PHASE_L2`	Terminal-block Parseval theorem	Phase L^2 is not a named atom
`TPC183.N_EQUALS_T`	Proposed terminal-to-cumulative specialization	Atom occurs only as a hypothesis
`O161.BAD_ENDPOINT_CONTRACT`	Verbal q/T cumulative all-prefix target	Prescribed atom has no locked value
`O161.DIRECT_TARGET_CONTRACT`	Verbal q/N target with no literal prefix	Prescribed atom has no locked value
`TW25.LOG_TWISTED_AFFINE`	Every fixed γ, logarithmically weighted	Fixed γ is not a production record
`PHYSICAL_PACKET_PREFIX`	Resolved unnormalized per-packet prefix	Symbolic $\alpha_{\xi,X}$ is not named production data

Table 1: The nine distinct plausible crosswalk objects. Native data are retained, but no row receives production credit by reinterpretation.

3 Three formula types

TPC-194 freezes the following three literal objects [9]:

$$\begin{aligned}
\mathcal{F}_N^{\mathrm{blk}}(\alpha) &= \frac{q}{N} \sum_{N<t(z)\le 2N} c_z \mathrm{e}(-\alpha z), && \texttt{CORE_TERMINAL_BLOCK},\\
\mathcal{A}_T(\rho) &= \frac{q}{T} \sum_{0<t(z)\le T} c_z \rho(z), && \texttt{CORE_CUMULATIVE_PREFIX},\\
S_{\xi,X}^{\le T} &= \sum_{\substack{z\in I_{\xi,X}\\ z\le T}} A_{\xi,X}(z)\mathrm{e}(-\alpha_{\xi,X} z), && \texttt{PHYSICAL_PACKET_PREFIX}.
\end{aligned}$$

The third object is unnormalized inside an outer physical packet sum. Setting $N = T$ in the first object gives $T < t(z) \le 2T$, not $0 < t(z) \le T$. No two rows can therefore be identified by renaming a parameter. A complete production crosswalk must give a literal directed bridge to each required object while retaining its domain, prefix index, and normalization.

4 The finite first-mismatch theorem

Theorem 1 (Locked registry first mismatch)**.** *In the declared TPC-204 corpus there are nine audited plausible crosswalk objects,* 63 *production-axis cells, and* 27 *formula-crosswalk cells. The number of complete production crosswalks is zero. For every candidate, the first failed production gate is*

$$\textit{NAMED\_PRODUCTION\_ATOM}.$$

Consequently the finite verdict is

$$\textit{FIRST\_MISMATCH\_CERTIFIED\_NOT\_TESTABLE}.$$

Proof. TPC-180 has zero value-bearing named-phase records and zero packet-coordinate rows, so its registry slot is empty [4]. The rational terminal-block theorem has a rational atom only conditionally on a supplied value and denominator; neither is a production registry value. TPC-159 uses an arbitrary periodic multiplier and excludes its dyadic shadow [2]. TPC-167 controls phase L^2, and its complete Fourier grid is an auxiliary averaging device rather than a packet schedule [3].

The TPC-183 specialization uses the atom, ranges, constant, and exponent only as hypotheses, and its $N = T$ substitution does not convert the terminal domain into the cumulative domain. TPC-184 freezes a verbal q/T cumulative all-prefix bad-endpoint target, but no source value for its prescribed atom or exact production schedule; it is not the shadow-excluding TPC-159 theorem. TPC-189 separately freezes a verbal direct q/N target but no source value for its prescribed atom [5, 6, 7]. The Teräväinen–Walker theorem controls every fixed γ, but its logarithmic affine average supplies neither a production atom record nor the natural target normalization [1, 8]. Finally, TPC-194's $\alpha_{\xi,X}$ is a symbolic per-key atom and $\xi = (\theta, c, \kappa, r)$ is a resolved key, not a production schedule.

Thus the first production-axis cell is false in every row. The remaining six axes and all three formula links are nevertheless audited: native ranges, constants, and normalizations are preserved with their original object and norm types, while logarithmic savings, phase averages, incomplete ledgers, and unattached normalizations receive no production credit. Hence no candidate satisfies the conjunction, and the asserted first mismatch follows. □

5 Scope, stop cell, and open parents

The exact new cell is

$$\texttt{TPC204\_DECLARED\_PLAUSIBLE\_PRODUCTION\_CROSSWALK\_CORPUS\_V1} = \texttt{STOP\_SCOPED}.$$

It stops only extraction of a complete crosswalk from this explicit finite corpus. It is not a theorem that no production registry exists in a larger source universe. TPC-193's V1 corpus remains `STOP_SCOPED`. Both `O161` pointwise parents and the global architecture remain open.

Even a future unique formula crosswalk would not itself prove a theorem-backed natural-q/N, named fixed-atom positive-X-power estimate. Here fixed-atom decay is false, named-atom endpoint credit is zero, the strict $1/400$ budget is unpaid, and $L2 = \texttt{NONE}$.

6 Machine certificate and trust boundary

The canonical payload records all nine candidates in a fixed order, all seven axis rows per candidate, all three formula links per candidate, four explicit exclusions, the source hashes, and the complete claim firewall. A separate read-only checker imports neither the builder nor the materialization contract. It independently freezes the nine candidate rows and revalidates TPC-194/203 hardening, exact TeX anchors and JSON fields, strict JSON types, the verdict truth table, twelve active-schema mutations, forty-five coordinated semantic mutations under regenerated schemas, and five additional nested Boolean/integer type-confusion attacks.

The manifest pins the machine artifacts, manuscript, bibliography, and stable PDF. It is a repository review pin, not an external signature or theorem source. Coordinated changes to code and manifest still require ordinary commit review.

References

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